

Geometry

Unit 8

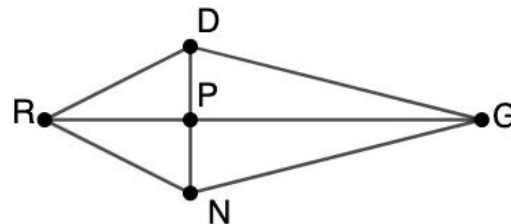
Lesson 4 Homework

Name **Answer Key** _____

Date _____

Period _____

Kite $RDGN$ is shown with diagonals $\overline{RG}$ and $\overline{DN}$ intersecting at point P where $DP = 13$ and $RP = 15$.



1. What is the length of $\overline{RD}$?

$$RD^2 = 13^2 + 15^2$$

$$RD^2 = 169 + 225$$

$$RD = \sqrt{394} \approx 19.849$$

2. Given $PG = 22$. What is the length of $\overline{NG}$?

$$NG^2 = 13^2 + 22^2$$

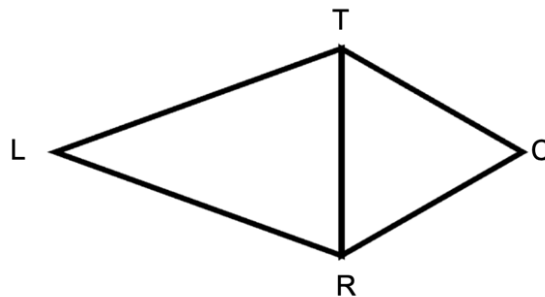
$$NG^2 = 169 + 484$$

$$NG = \sqrt{653} \approx 25.554$$

3. What is the perimeter of $RDGN$?

$$\text{Perimeter} = 2 \cdot 19.849 + 2 \cdot 25.554 = 90.806$$

$TCRL$ is a kite as shown where $m\angle CTR = (7x - 12)^\circ$, $m\angle TCR = (8x + 32)^\circ$, $m\angle LRT = (9y + 40)^\circ$, and $m\angle TLR = (4y)^\circ$.



4. What is the value of x ? Round your answer to the nearest hundredth.

$$8x + 32 + 2(7x - 12) = 180$$

$$8x + 32 + 14x - 24 = 180$$

$$22x + 8 = 180$$

$$x = 7.82$$

5. What is the value of y ? Round your answer to the nearest hundredth.

$$4y + 2(9y + 40) = 180$$

$$4y + 18y + 80 = 180$$

$$22y = 100$$

$$y = 4.55$$

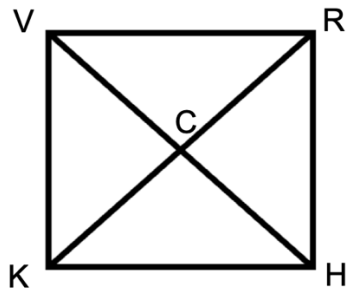
6. What is the measure of $\angle CTL$?

$$m\angle CTL = m\angle CTR + m\angle RTL$$

$$m\angle CTL = 42.74 + 80.95$$

$$m\angle CTL = 123.69^\circ$$

Quadrilateral $VRHK$ is shown with diagonals $\overline{VH}$ and $\overline{RK}$ intersecting at point C .



7. What must be true for $VRHK$ to be a square?

All sides are congruent
All angles are congruent
Diagonals are congruent
Diagonals bisect each other

8. Given $VRHK$ is a square with $VC = 6z$, and $VR = 12$. What is the value of z ? Leave your answer in radical form.

$$\begin{aligned}(6z)^2 + (6z)^2 &= 12^2 \\ 36z^2 + 36z^2 &= 144 \\ 72z^2 &= 144 \\ z &= \sqrt{2}\end{aligned}$$

9. What is the length of $\overline{RK}$, in simplest radical form?

$$RK = 6\sqrt{2} + 6\sqrt{2} = 12\sqrt{2}$$

10. What is the length of $\overline{VK}$, in simplest radical form?

$$VK = 12$$